Stochastic Simulation of Virus Population Dynamics

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Introduction

Viruses may become resistant to one drug, sometimes to multiple drugs due to mutations.

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- Viruses such as HIV, Influenza, and Ebola are difficult to treat because of their ability to evolve extremely quick.
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- Computational modeling has played an important role in the study of viruses such as HIV, by simulating genetic errors in the genome replication process.
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- Analyzing the medication for the treatment of infection by virus under different conditions
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- Observation of the response of virus population to the introduction of drugs.

Simple Simulation (No Drug Treatment)

Modeling of Virus Population without Drug Administration

- No intake of drug by the patient.
- Viruses do not acquire resistance to drugs.
- Virus population in a patient assumed to be left untreated.
- At every time step fixed probability of virus particle of being cleared from patient's body.
- Uncleared virus particle considered for reproduction.
- Probability of virus particle reproduction is a function of virus population.

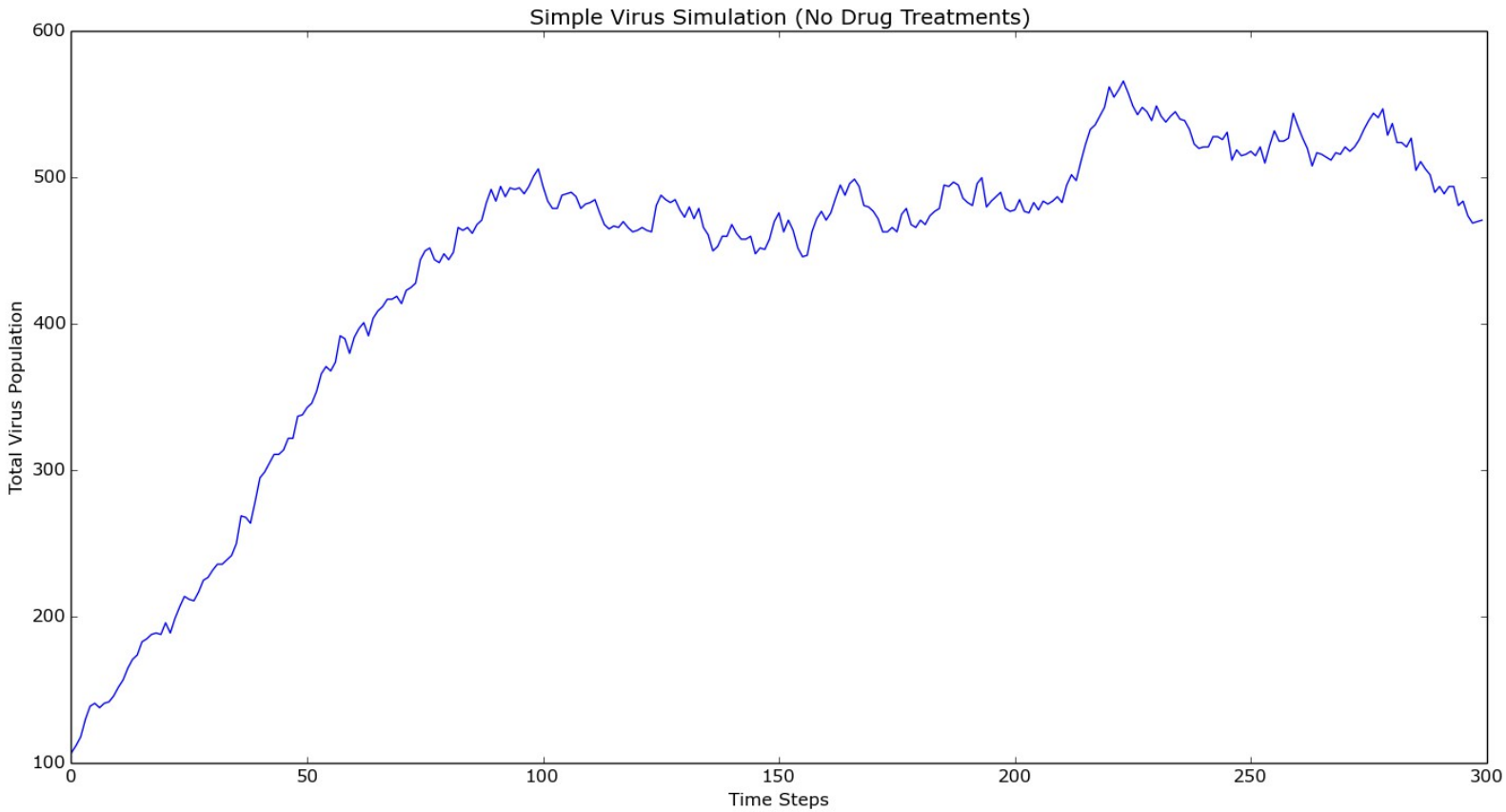
Parameters used in simulation

- Initial number of identical viruses in patient = 100
- Maximum Sustainable Virus Population = 1000
- Maximum Reproduction Probability for a Virus Particle = 0.1
- Maximum Clearance Probability for a Virus Particle = 0.05
- Number of time steps used in simulation = 300

Limitations

- Virus population induces fewer resources in the patient's body to facilitate and lowering the probability reproduction.
- not reproduce on their own, make use of patient's cells to reproduce.

Analyzing a Simple Simulation (No Drugs Treatments)



Inferences:

- After about 100 time steps, population stops growing and fluctuates irregularly between 475 and 550.
- As the virus population increases, there will be fewer available host cells for viruses to utilize for reproduction.

Simulation with Drugs

Modeling of Virus Population considering the effects of

- Administering drugs to the patient
- Ability of virus particle offspring to inherit
- Mutation genetic traits that confer drug resistance

Conditions

- Virus population reproduces, mutations occur in the virus offspring and adds to the genetic diversity to the virus population.
- Gain of favorable mutations that confer resistance to drugs
- Drugs do not kill directly virus particles lacking resistant to drugs, prevent them from reproducing.
- Virus particles with resistance to drug continue to reproduce normally.

Parameters used in simulation

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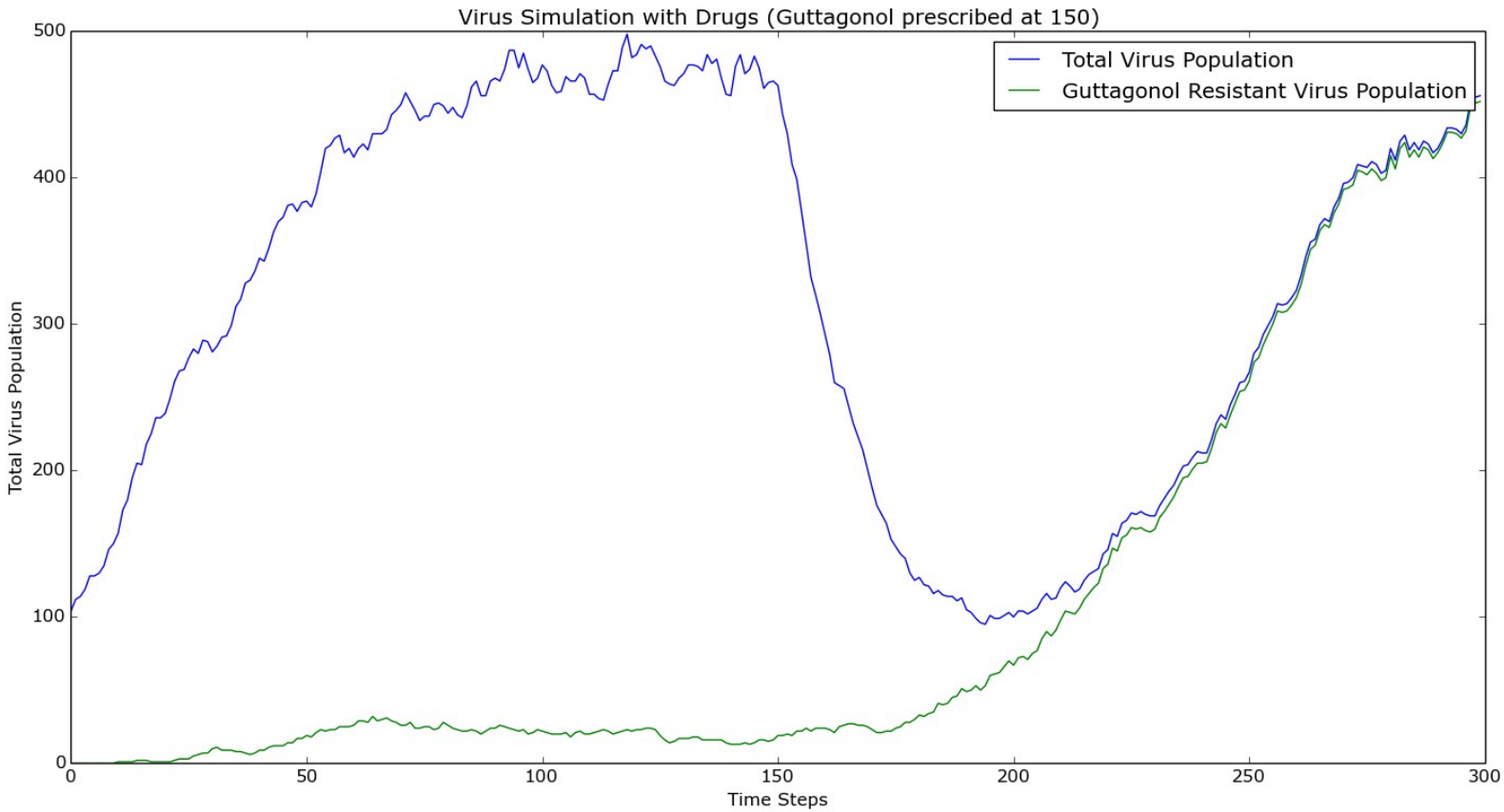
Analyzing Simulation with Single Drug



Inferences:

- Virus population drops after the introduction of Guttagonol prescribed at 150th time step.
- Given sufficient time, and if the administered drug is weak, the virus undergoes mutation and develops resistance against Guttagonol.

The Effect of Delaying Treatment on Patient Outcome

Modeling of Virus Population considering the effects of delayed treatment

- Explore the effect of delaying treatment on the ability of the drug to eradicate the Virus population.
- Run multiple simulations to observe trends in the distributions of patient outcomes.
- Simulation is run for 300, 150, 75, and 0 time steps before administering Guttagonol to the patient.
- The simulation is run for an additional 150 time steps after administering Guttagonol.
- Repetition of the experiment multiple times, with recording of the final virus population.

Parameters used in simulation

- Initial number of identical viruses in patient = 100
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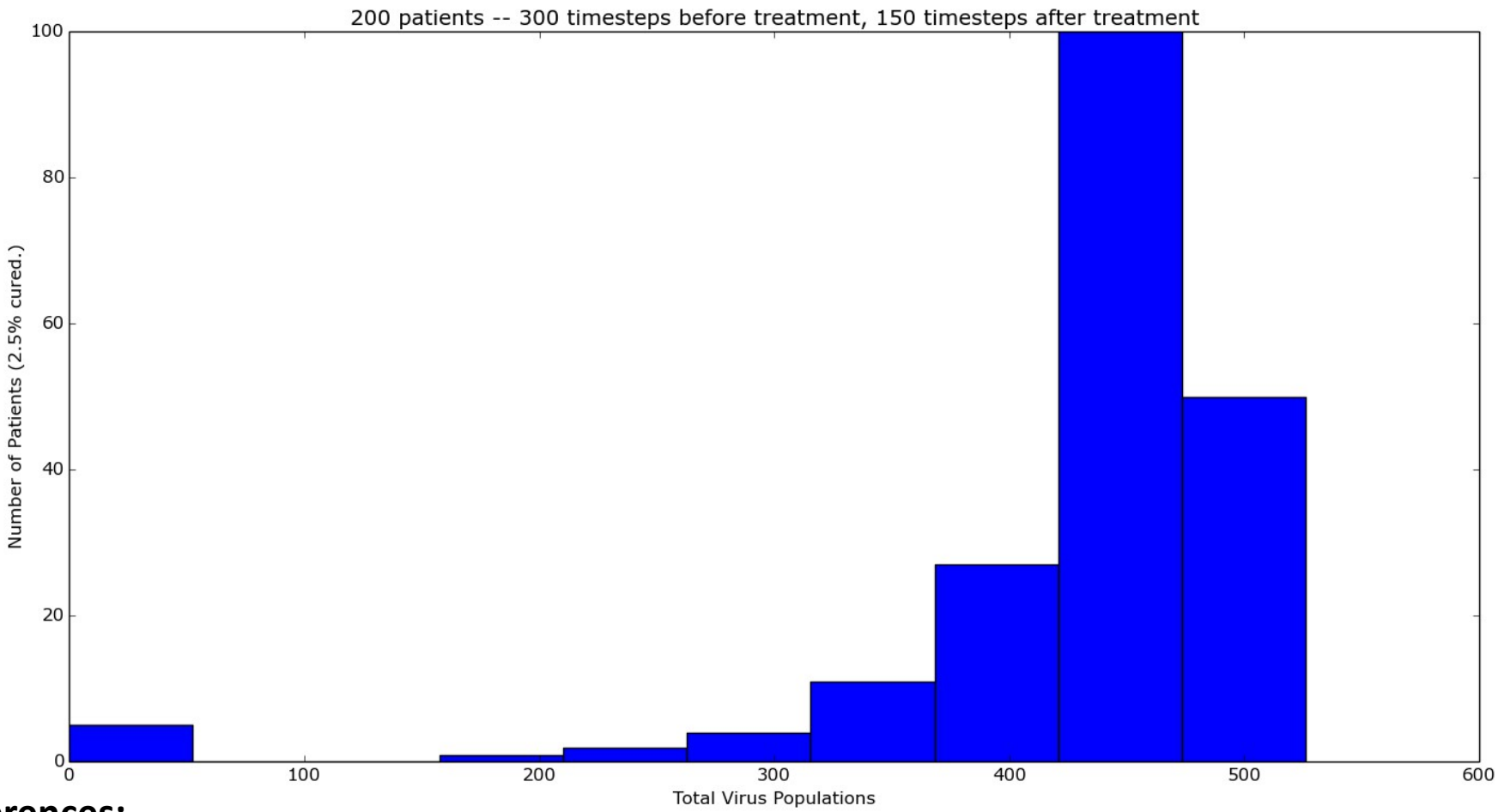
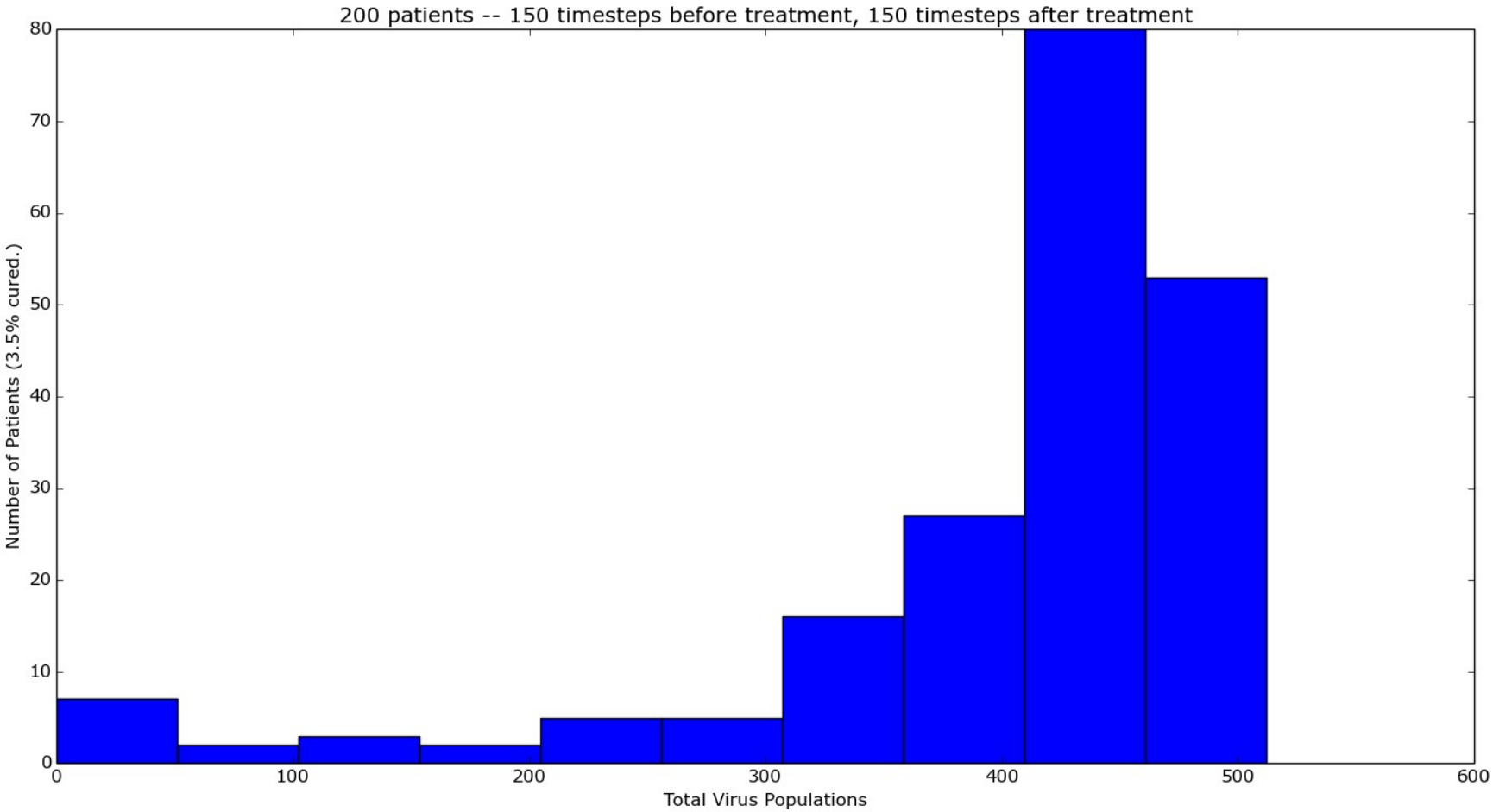
Analyzing the Effect of Delayed Treatment



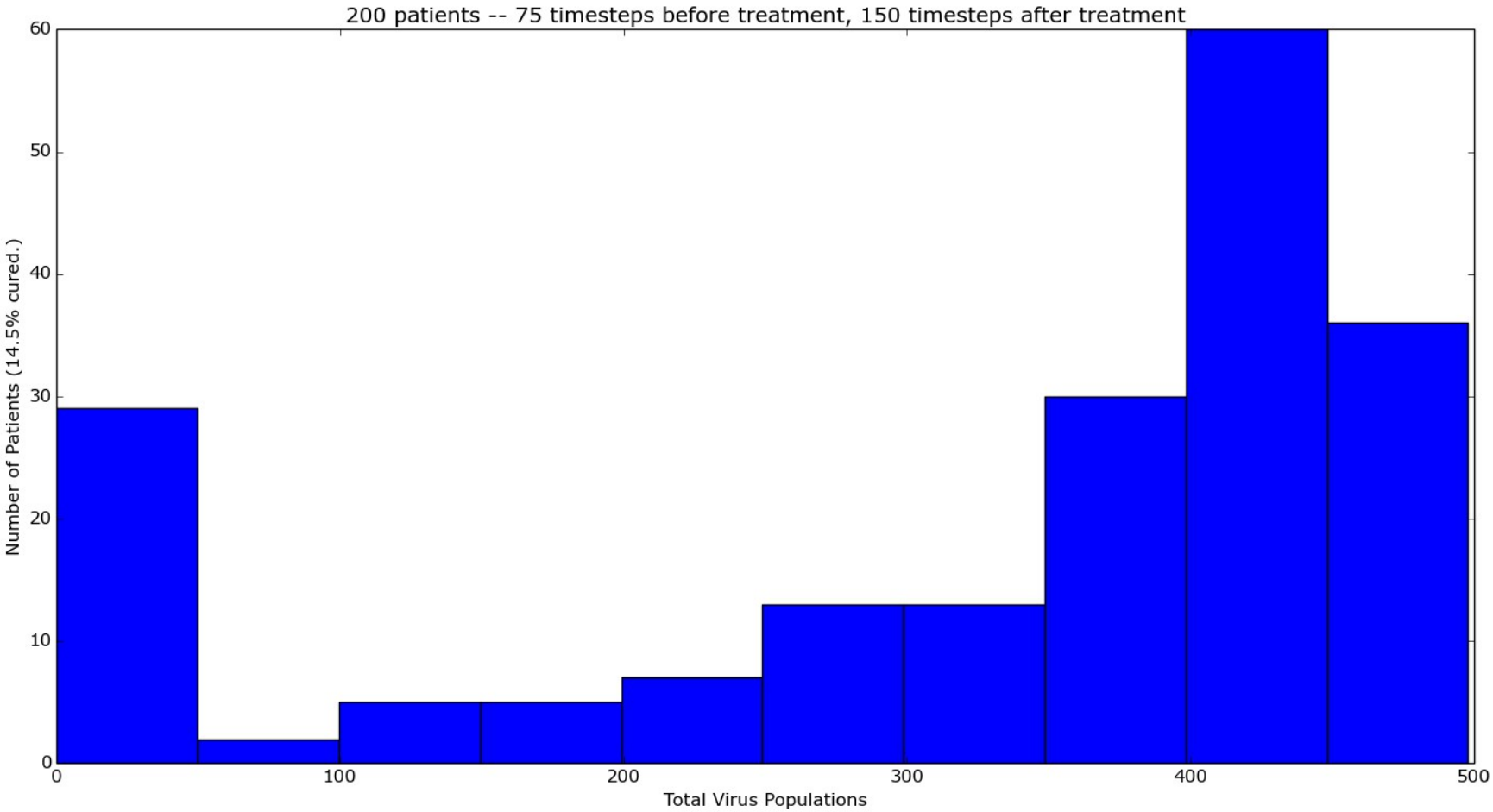
Inferences:

- Since the treatment was started very late, majority of the patients had a huge amount of virus population in their body.
- 2.5% out of 200 patients had virus population less than 50, and were considered cured.

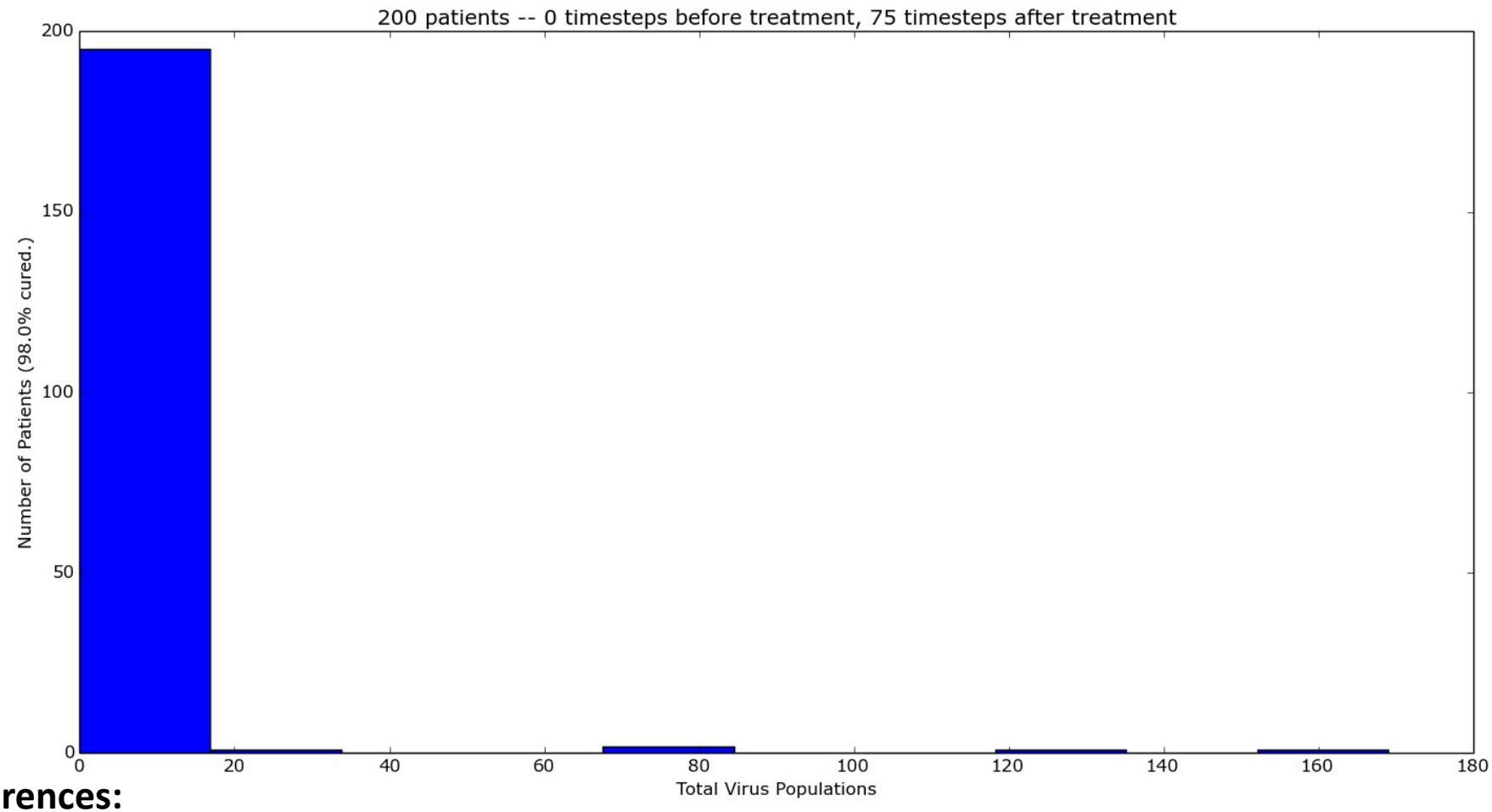
Analyzing the Effect of Delayed Treatment



Analyzing the Effect of Delayed Treatment



Analyzing the Effect of Delayed Treatment



Inferences:

- Immediate start of treatment resulted in curing of the disease in 98% of the cases.

Conclusion

- Virus population is an increasing function till some definite time step, after which it fluctuates within a range.
- Given sufficient time, and depending upon the concentration of the drug, a virus can mutate and develop resistance against the drug, after which it has no effect on the virus.
- Delaying the treatment of a virus on a patient reduces the chances of him/her getting cured.
- Combined effect of two drugs prevents the virus to develop resistance against either of the drugs.
- Non-compliance of a patient, when he/she refuses to take drugs regularly gives more time to the virus population to gain resistance against the drug.

Thank You